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HYDRAS

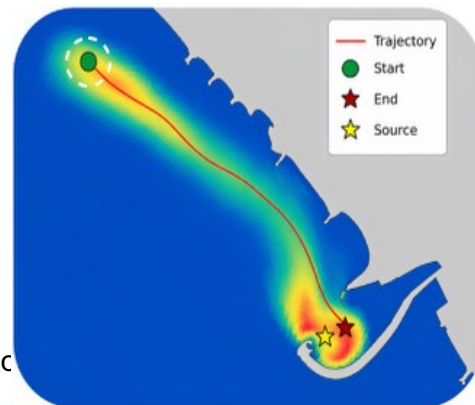
HYdrodynamic-aware Distributed Robots for mArine Source-seeking



Marine source seeking

Guide autonomous marine vehicles toward an unknown pollutant source using pointwise concentration measurements

- **Local sensing in space and time:** each sensor measures concentration at its own location at discrete sampling times, without observing the full field or the source position
- **Dynamic concentration field:** advection and diffusion redistribute the pollutant, so the local concentration gradient may not point toward the source
- **Motion guided by measurements:** use successive observations to decide where to sample next and steer the vehicles toward the source as the field evolves



Source-seeking methods: main approaches

MODEL-BASED

- Transport-model inversion
- Bayesian source estimation
- Information-driven planning
- ✓ Use source estimates and uncertainty to guide motion
- X Sensitive to model mismatch; may require substantial online computation

Zhang, Qin, Tang, Li, "Distributed information-based source seeking," *IEEE Trans. on Robotics*, 2023

MODEL-FREE

Training-free

- Field climbing
- Extremum seeking
- Bio-inspired plume tracing
- ✓ No offline training; typically low online computational cost
- X Rely on predefined search rules; gradient-based search can stall in weak gradients or converge to local concentration maxima

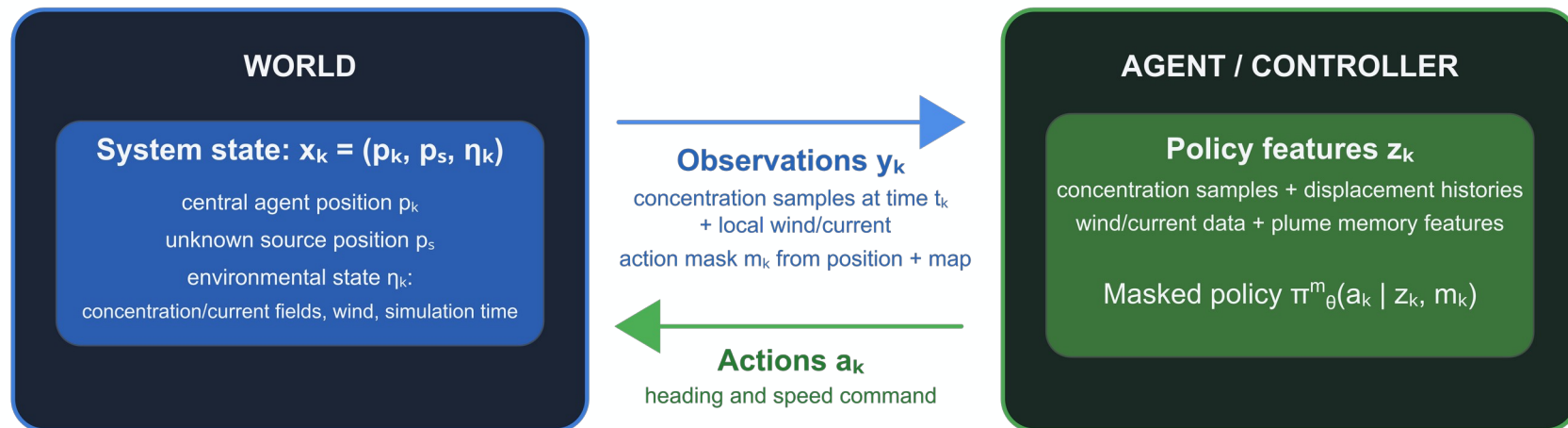
Briñón-Arranz, Renzaglia, Schenato, "Multirobot Symmetric Formations for Gradient and Hessian Estimation With Application to Source Seeking," *IEEE Trans. on Robotics*, 2019

Reinforcement Learning

- **Policy:** maps observations and memory to motion commands
- **Training:** learns the policy through interaction with simulated world to maximize cumulative reward
- ✓ Fast online action selection; no transport-model solution
- X Depends on reward design and training coverage; generalization to unseen conditions requires validation

Hu, Song, Chen, "Plume Tracing via Model-Free Reinforcement Learning Method," *IEEE Trans. on Neural Networks and Learning Systems*, 2019

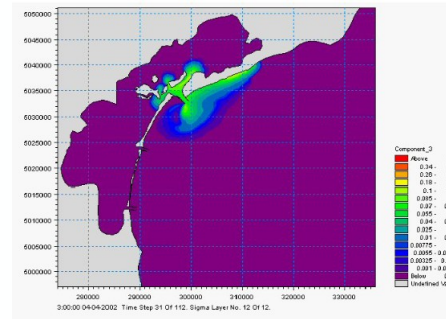
Source seeking as a partially observable Markov decision process



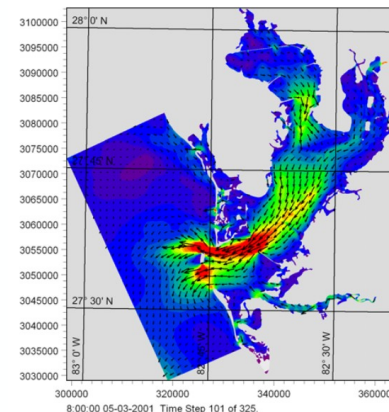
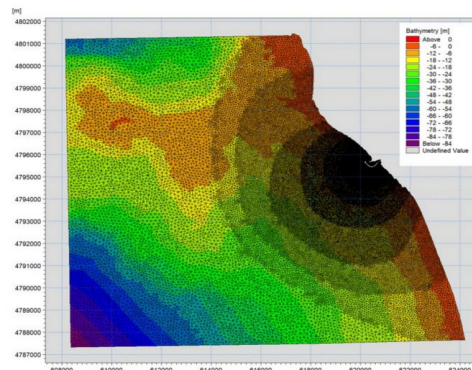
- **Actions:** select the central agent's heading and speed from admissible commands
- **Objective:** reach a neighbourhood of the unknown source as quickly as possible using local measurements
- ✓ **Offline training:** learn a feedback policy through interaction with high-fidelity hydrodynamic simulations
- ✓ **Online execution:** use local measurements and memory to select admissible motion commands, without solving transport equations or explicitly estimating the source position

Dataset: hydrodynamic and transport modelling

- **Pollutant concentration fields from MIKE21 Flow Model FM (DHI):** 2D shallow-water equations, Eulerian advection-diffusion of a non-reactive tracer
- **Domain: Cecina bay** ($\sim 16 \times 14$ km), triangular mesh with variable resolution (~ 10 m near the port, ~ 150 m offshore)
- **Hydrodynamic downscaling:** from Mediterranean basin (~ 4.5 km) to local coastal mesh (~ 10 m)
- **Environmental forcing:** CMEMS currents and sea level at open boundaries (Flather conditions), plus time-varying, spatially uniform ERA5 wind
- **Reference hydrodynamic regime:** Class 2 (moderate currents, persistent direction and low sea-level variability, selected from four clusters of CMEMS data)



Tracer simulation of single component from outlet in the Adriatic, simulated by MIKE 21 Flow Model FM HD+TR

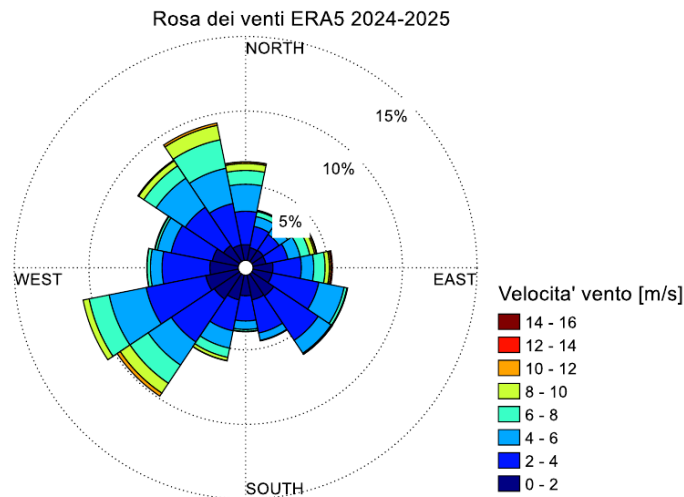


Example of a flow field in Tampa Bay, FL, simulated by MIKE 21 Flow Model FM



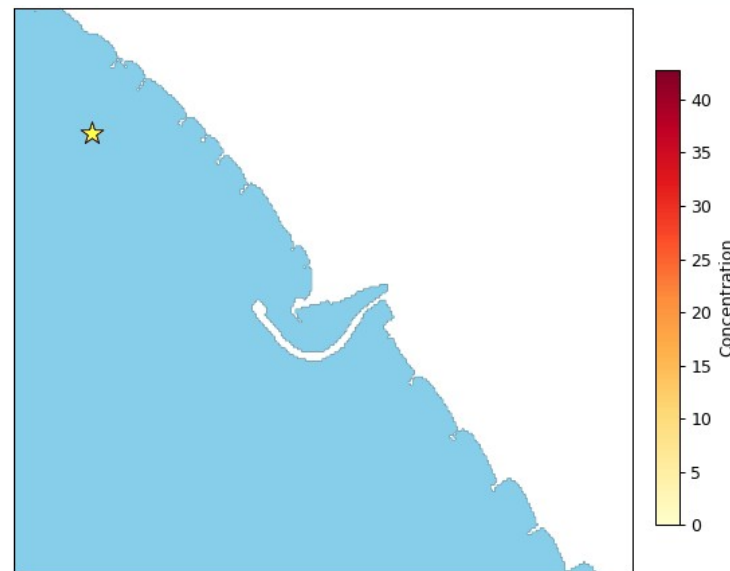
Dataset: wind scenarios

- **Class V0:** calm/light wind conditions; low-intensity wind (approximately 1.5-2 m/s), with no significant prevailing direction
- **Class V1:** persistent wind from the NW; prevailing direction between 310° and 330°, with a typical duration of 8-12 hours and speeds up to 8-10 m/s
- **Class V2:** persistent wind from the SW; prevailing direction between 230° and 250°, with a typical duration of 8-12 hours and speeds up to 8-10 m/s
- **Class V3:** wind from the eastern sector; prevailing direction between 110° and 130°, with a typical duration of 8-12 hours and speeds up to 6-8 m/s

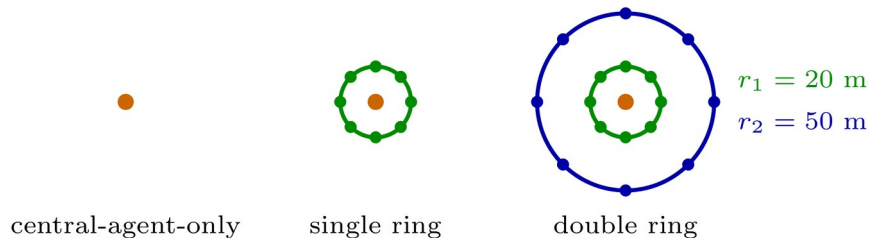


Dataset: training and evaluation setup

- **Dataset format:** NetCDF concentration and current fields on a 10 m grid, sampled every 2 min over a 3 × 2.5 km coastal window
- **Source scenarios:** 132 source locations (~200 m spacing), each simulated under four wind scenarios with identical, constant release conditions (50 L/s)
- **Training and evaluation:** 106 source locations for training and 26 held-out locations for evaluation
- **Initial plume age:** episodes start at 25%, 50% or 75% of the simulated time series, exposing the agent to different stages of plume evolution
 - Q1/4 (compact)
 - Q1/2 (partly dispersed)
 - Q3/4 (heavily dispersed)

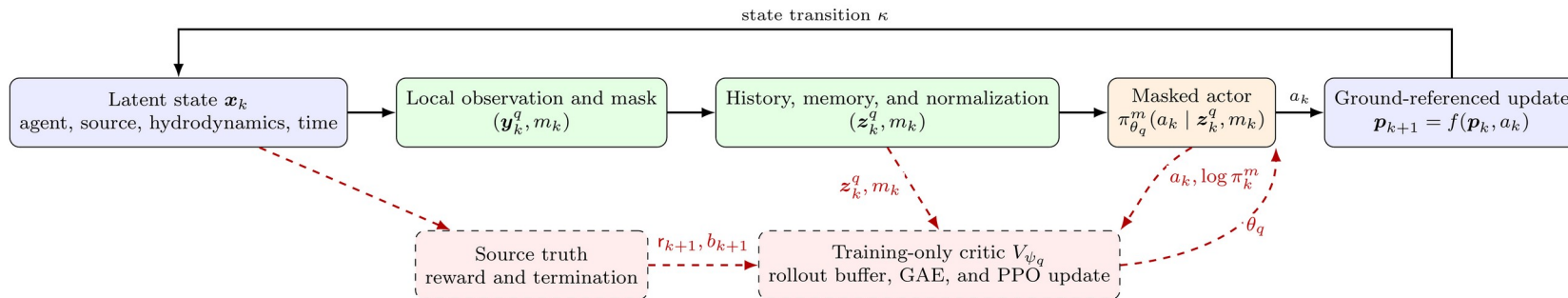


Centralized fixed-formation sensing



- **Centralized control:** one policy selects the central agent's action using all available measurements
- **Three sensing configurations:** central agent only; one ring with 8 additional sampling points at 20 m; two rings with 8 further points at 50 m
- **Ideal co-moving sensors:** ring locations maintain fixed offsets from the central agent, measurements instantaneously available to the policy
- **Single-agent decision process:** ring sensors have no independent policies or vehicle dynamics; communication constraints are not modeled

Policy architecture and training: Maskable PPO



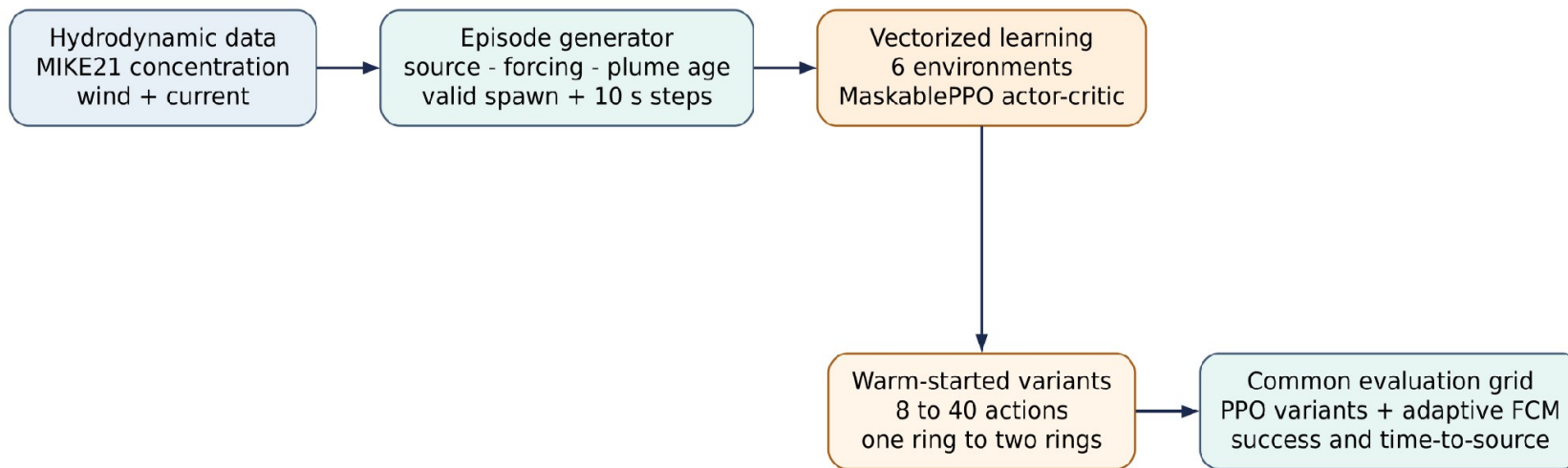
Architecture

- **Policy input:** 36/116/196 features for central-only/single-ring/double-ring sensing; current + 9 previous measurements
- **Actor-critic networks:** separate MLPs, each with two 256-unit hidden layers and tanh activations
- **Action space:** 40 heading-speed combinations (8 headings $\times$ 5 speed levels)
- **Action masking:** exclude commands whose endpoints lie on land or outside the domain before selecting an action

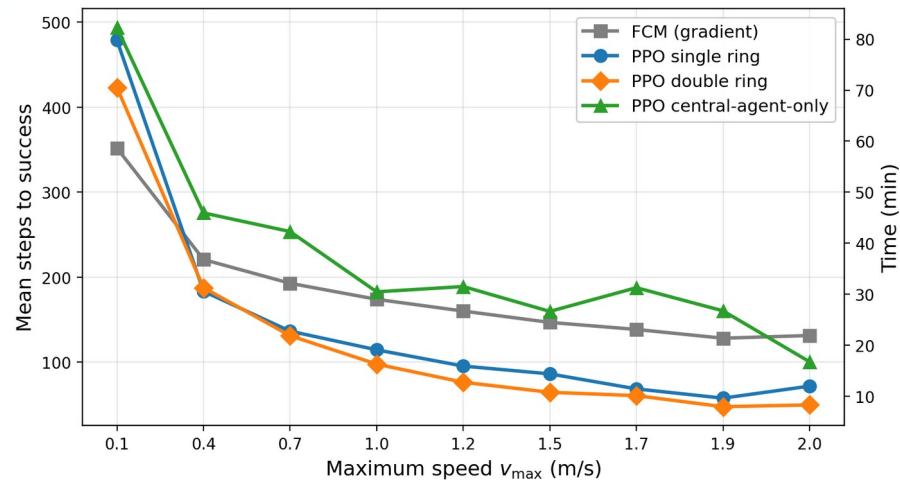
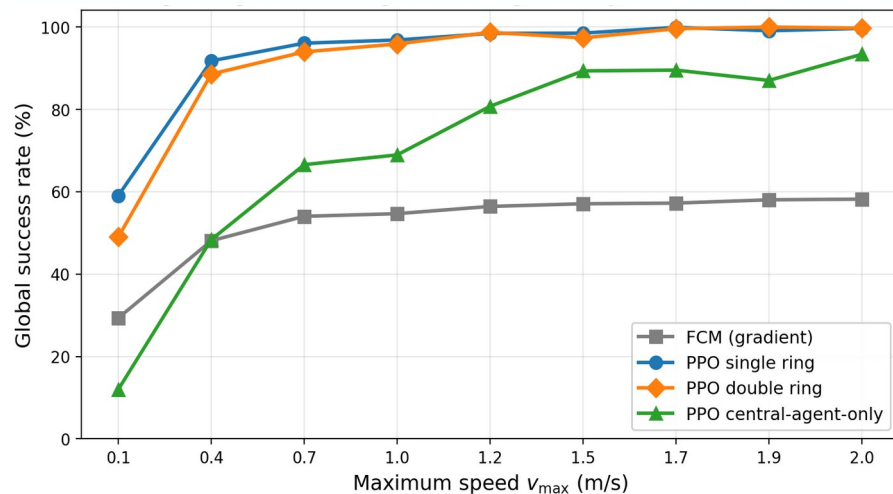
Training

- **Masked PPO:** clipped policy updates with Generalized Advantage Estimation (GAE)
- **Parallel sampling:** 6 environments $\times$ 4,096 steps = 24576 transitions per rollout
- **Optimization:** minibatches of 128 transitions, up to 10 epochs per update, clipping parameter $\epsilon = 0.15$

Hydrodynamic data flow



Experimental results



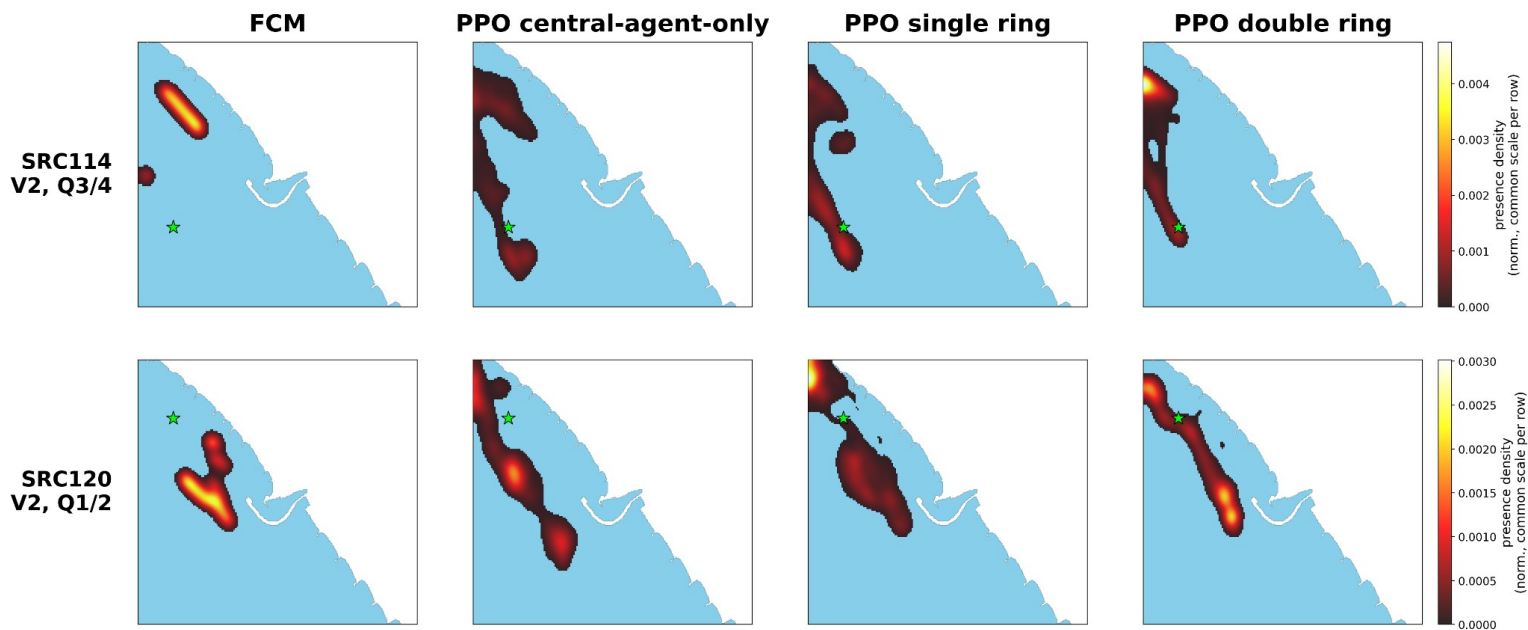
Experimental results

Model	SR%	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
FCM Adam ($lr=12$)	56.4	76.3	29.5	37.2	82.7	79.8	34.6	54.8
PPO, central-agent-only	79.0	86.9	86.7	64.9	77.7	91.0	68.5	77.7
PPO single ring	98.4	99.7	100.0	95.6	98.2	100.0	95.8	99.4
PPO double ring	98.7	99.2	100.0	95.4	100.0	100.0	99.6	96.3

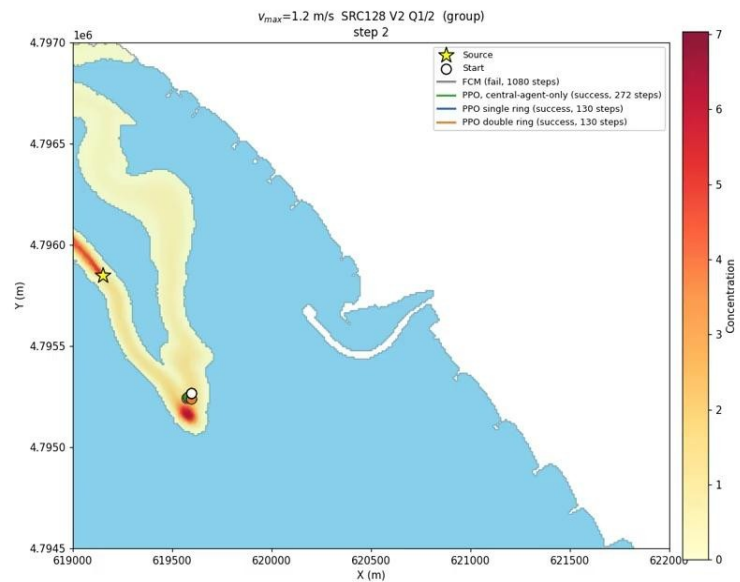
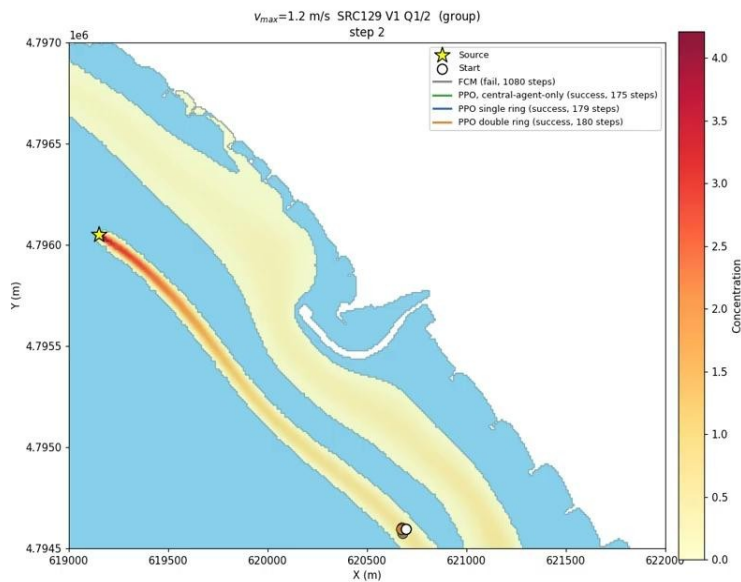
- Total held out sources: 26
- Total episodes: $26 \times 4 \text{ winds (V}^*) \times 3 \text{ plume ages (Q}^*) \times 5 \text{ episodes for combination} = 1560 \text{ episodes}$

Experimental results

Occupancy heatmaps — $v_{max} = 1.2$ m/s (★ = source, white = land, blue = sea)



Video demonstration

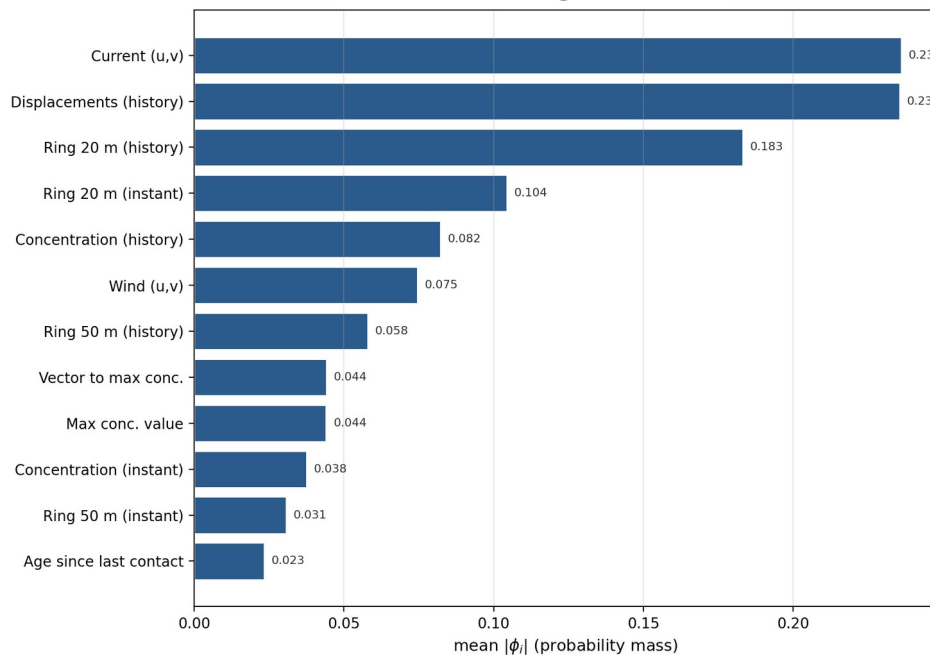


Conclusions

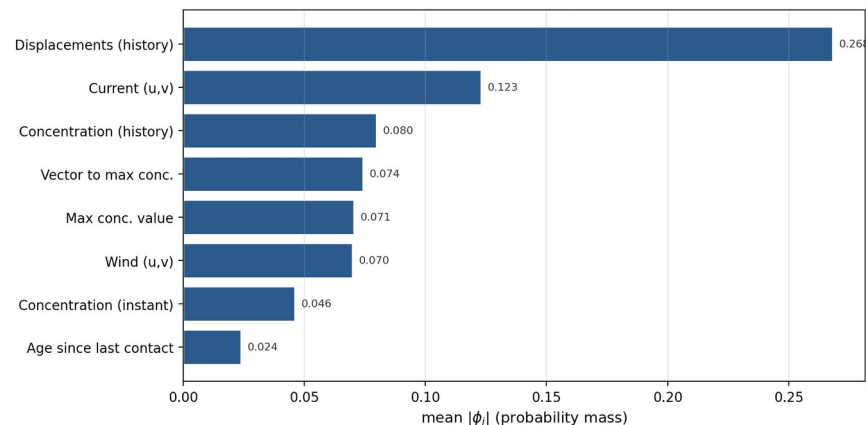
- **Does RL outperform classical field climbing?**
Yes. At 1.2 m/s, ring-based PPO achieves **98.4-98.7% success** versus **56.4%** for FCM. FCM performs poorly in diffuse-plume conditions (**V1**, **V2** and **Q1/2**), and representative trajectories show trapping at local concentration maxima away from the source.
- **Does spatial sensing improve performance?**
Yes, across the evaluated speed range (0.1-2.0 m/s). Ring-based policies achieve higher overall success rates and fewer mean successful steps than central-agent-only PPO. At 1.2 m/s, the outer ring reduces mean successful steps from **95 to 75**, with similar success rates.
- **Which reward works best?**
The minimal variant achieves the highest success rate in the reward comparison: **96.9%**, without explicit wind/current alignment bonuses.
- **How does the policy use the available information?**
It combines **concentration** and **motion histories** with **local current information**. Ablations show the strongest impact on success from ring-measurement history with formation and displacement history without formation.

Policy interpretation: group Shapley analysis

Double-ring PPO



Central-agent-only PPO



Conclusions

Limitations

- **Incomplete mission:** plume detection is assumed; source arrival is verified using ground truth
- **Centralized decision-making:** sensing rings have no independent policies or vehicle dynamics
- **Limited temporal representation:** fixed observation history; remaining episode time is not provided
- **Idealized deployment:** simplified motion, noise-free sensing, ideal communication; missing readings are zero-filled
- **Limited environmental coverage:** one coastal domain and one representative boundary-forcing regime

Future work

- **End-to-end autonomy:** exploration, plume detection, source seeking, onboard arrival detection
- **Cooperative multi-agent control:** decentralized policies and information sharing, e.g. MAPPO/CTDE
- **Time-aware policies:** remaining-time input and learned memory using LSTM or Transformer models
- **Realistic deployment and validation:** vehicle dynamics, sensor noise and delays, domain randomization
- **Generalization across hydrodynamic regimes and coastal domains:** train under diverse hydrodynamic conditions, assess generalization to unseen environments



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Thank you for your attention